KINGSCLIFF HIGH SCHOOL
ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 9 STANDARD

CHAPTER 10

YEAR 9 MATHEMATICS

Surface area and volume

10 lessons · three pages each · Easy drill, then Medium and worded problems

10.1 Metric units
10.2 Precision and absolute error
10.3 Percentage error
10.4 Perimeter and area of composite shapes
10.5 Areas of quadrilaterals
10.6 Circumference and arc length
10.7 Areas of circles and sectors
10.8 Surface area of a prism
10.9 Surface area of a cylinder
10.10 Volumes of prisms and cylinders

NAME _____ CLASS _____ TEACHER _____

Artwork: a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver, part of the Mandelbrot set.

How to use this booklet

1. Each lesson is **three pages**. Read the **learning intention**, copy your teacher's notes into the **Notes** box, and tick each **success criterion** when you can do it.

2. Pages 1 and 2 are **Easy**: small steps, one round at a time. Page 3 is **Medium**, then three **worded problems**.

WE DO **Grey boxes.** Your teacher works each example on the board. **Copy every step** into the space.

YOU DO **Blue boxes.** Your turn. The **?** on each diagram shows what to find. Write the **formula**, **substitute** the numbers, then the **answer** with its unit.

3. Some Medium questions come from earlier lessons. The tag (like **10.2** or **Year 8**) says which one.

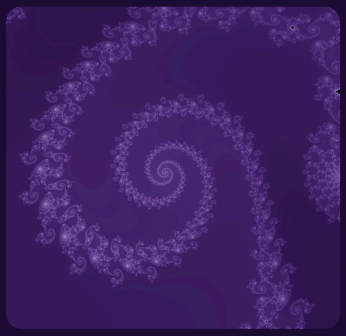
4. Stuck? Read the **Remember** line in the blue box, then look back at your notes and the example. Answers with π: **one decimal place** unless the question says otherwise.

Contents and progress tracker

Lesson	Topic	Page	Date	Done
10.1	Metric units	3		☐
10.2	Precision and absolute error	6		☐
10.3	Percentage error	9		☐
10.4	Perimeter and area of composite shapes	12		☐
10.5	Areas of quadrilaterals	15		☐
10.6	Circumference and arc length	18		☐
10.7	Areas of circles and sectors	21		☐
10.8	Surface area of a prism	24		☐
10.9	Surface area of a cylinder	27		☐
10.10	Volumes of prisms and cylinders	30		☐

In this chapter you will

- convert between metric units, including very large and very small units
- find the precision, absolute error and percentage error of a measurement
- find the perimeter and area of composite shapes and quadrilaterals
- find the circumference, arc length and area of circles, sectors and composite shapes
- find the surface area of prisms and cylinders
- find the volume and capacity of prisms and cylinders



About the artwork

The pictures in this booklet are parts of the **Mandelbrot set**, drawn by a computer that repeats one simple rule, $z \to z^2 + c$, thousands of times for every point. Every booklet shows a different part of the set: this one is a silver spiral on the western edge of Seahorse Valley, coloured in purple and silver. Maths can make beautiful things.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

10.1 Metric units

EASY

Learning intention We are learning to convert between metric units, including time and digital storage.

Success criteria
- ☐ I can convert units of length, mass and capacity
- ☐ I can convert units of time and digital storage
- ☐ I can use the prefixes mega-, giga- and micro-

NOTES

Copy what your teacher writes on the board.

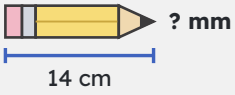
Date _____

WE DO

With your teacher: copy what they write on the board.

Example 1 Convert each one. To a smaller unit, multiply. To a bigger unit, divide.

a Length



b Length



c Mass



d Capacity



YOU DO

On your own: work down the box, one round at a time. Show your working.

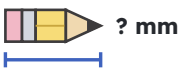
8 QUESTIONS

REMEMBER To a smaller unit, multiply · To a bigger unit, divide · 1 km = 1000 m · 1 m = 100 cm · 1 cm = 10 mm

ROUND 1

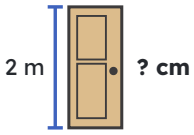
Questions 1–4: convert each length to the smaller unit.

1



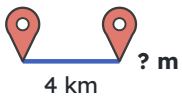
Rule × 10
Working
Answer

2



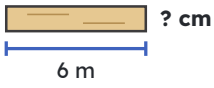
Rule × 100
Working
Answer

3



Rule × 1000
Working
Answer

4

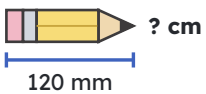


Rule × 100
Working
Answer

ROUND 2

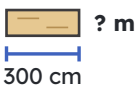
Questions 5–8: convert each length to the bigger unit.

5



Rule ÷ 10
Working
Answer

6



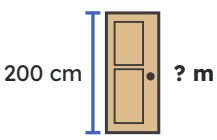
Rule ÷ 100
Working
Answer

7



Rule ÷ 1000
Working
Answer

8



Rule ÷ 100
Working
Answer

10.1 Metric units

EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER 1 kg = 1000 g · 1 L = 1000 mL · 1 min = 60 s · 1 h = 60 min · To a smaller unit, ×. To a bigger unit, ÷.

ROUND 3 Questions 9–11: convert each mass.

9



Rule × 1000
Working
Answer

10



Rule ÷ 1000
Working
Answer

11



Rule × 1000
Working
Answer

ROUND 4 Questions 12–14: convert each capacity.

12



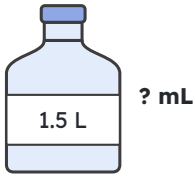
Rule × 1000
Working
Answer

13



Rule ÷ 1000
Working
Answer

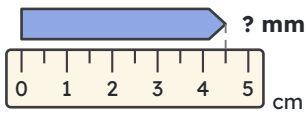
14



Rule × 1000
Working
Answer

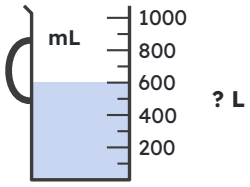
ROUND 5 Questions 15–17: × or ÷? Write the rule, then convert. Read the scales carefully.

15



Rule
Working
Answer

16



Rule
Working
Answer

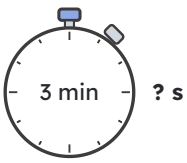
17



Rule
Working
Answer

ROUND 6 Questions 18–20: convert each time. 1 min = 60 s and 1 h = 60 min.

18



Rule × 60
Working
Answer

19



Rule × 60
Working
Answer

20



Rule ÷ 60
Working
Answer

10.1 Metric units

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Convert each one. Write the rule first.**

a Time



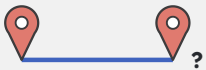
? min

b Storage



? MB

c Two steps



? cm

d Micro



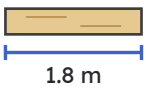
? µm



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

REMEMBER 1 GB = 1000 MB · 1 ML = 1 000 000 L · 1 g = 1000 mg · 1 mm = 1000 µm · m → mm: × 100, then × 10

1 Length



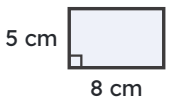
? mm

Rule _____

Working _____

Answer _____

2 Area Year 8

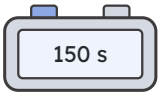


Formula _____

Substitute _____

Answer _____

3 Time



? min

Rule _____

Working _____

Answer _____

4 Round 4.0862 m to two decimal places. Year 8

Answer _____

5 Storage



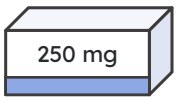
? MB

Rule _____

Working _____

Answer _____

6 Mass



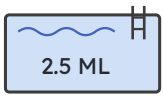
? g

Rule _____

Working _____

Answer _____

7 Capacity



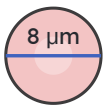
? L

Rule _____

Working _____

Answer _____

8 Length



? mm

Rule _____

Working _____

Answer _____

9 Find 15% of 60 kg. Year 8

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A phone has 64 GB of free space. Each video is 800 MB. How many videos fit on the phone?

Answer _____

2 Mia runs laps of a 400 m track. How many laps does she run to cover 6 km?

Answer _____

3 A movie runs for 135 minutes. It starts at 7:30 pm. At what time does it finish?

Answer _____

10.2 Precision and absolute error

EASY

Learning intention We are learning to find the precision, absolute error and limits of accuracy.

Success criteria
- ☐ I can find the precision of a scale or a measurement
- ☐ I can find the absolute error: precision ÷ 2
- ☐ I can write the limits of accuracy: m ± absolute error

NOTES

Copy what your teacher writes on the board.

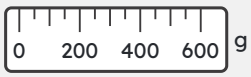
Date _____

WE DO

With your teacher: copy what they write on the board.

Example 1 Find the precision, the absolute error or the limits of accuracy.

a Precision



b Abs. error



c Abs. error



d Limits



YOU DO

On your own: work down the box, one round at a time. Show your working.

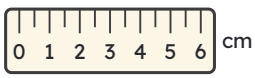
6 QUESTIONS

REMEMBER p = jump between two numbers ÷ number of spaces between them · Abs. error = p ÷ 2

ROUND 1

Questions 1–3: find the precision of each scale: the size of one small space.

1



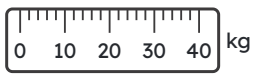
precision = ?

Formula: p = jump ÷ spaces

Substitute: _____

Answer: _____

2



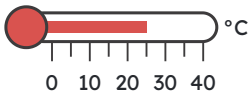
precision = ?

Formula: p = jump ÷ spaces

Substitute: _____

Answer: _____

3



precision = ?

Formula: p = jump ÷ spaces

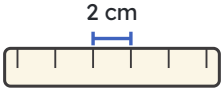
Substitute: _____

Answer: _____

ROUND 2

Questions 4–6: find the absolute error. The precision (one small space) is given.

4



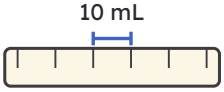
abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute: _____

Answer: _____

5



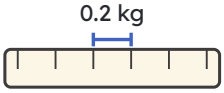
abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute: _____

Answer: _____

6



abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute: _____

Answer: _____

10.2 Precision and absolute error

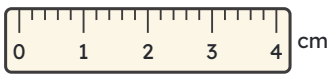
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER p = jump ÷ spaces · Abs. error = p ÷ 2 · p of 3.45 m = 0.01 m (its last digit) · Limits = m ± abs. error

ROUND 3 Questions 7–9: find the absolute error of each scale.

7



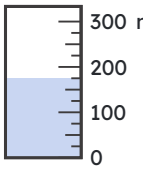
abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute:

Answer:

8



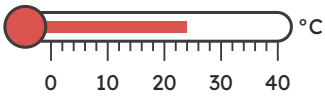
abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute:

Answer:

9



abs. error = ?

Formula: Abs. error = p ÷ 2

Substitute:

Answer:

ROUND 4 Questions 10–12: write the formula, then find the absolute error. The last digit gives p.

10



3.45 m

abs. error = ?

Formula:

Substitute:

Answer:

11



abs. error = ?

Formula:

Substitute:

Answer:

12



abs. error = ?

Formula:

Substitute:

Answer:

ROUND 5 Questions 13–15: find the limits of accuracy of each measurement.

13



6 m

limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

14



limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

15



limits = ?

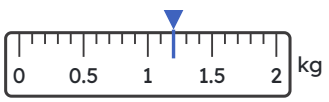
Formula: Limits = m ± abs. error

Substitute:

Answer:

ROUND 6 Questions 16–18: read each scale, then find the limits of accuracy.

16



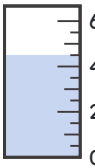
limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

17



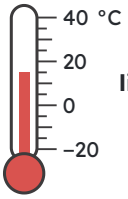
limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

18



limits = ?

Formula: Limits = m ± abs. error

Substitute:

Answer:

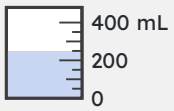
10.2 Precision and absolute error

MEDIUM

WE DO With your teacher: copy what they write on the board.

Example 2 **Find the limits of accuracy of each measurement.**

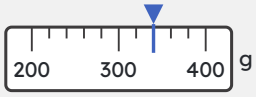
a Limits



b Limits



c Limits



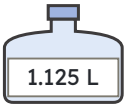
d Could it be 3.46 kg?



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

REMEMBER Lower limit = m − abs. error · Upper limit = m + abs. error · The true value is between the limits

1 Abs. error



Formula _____

Substitute _____

Answer _____

2 Convert 10.1



= ? mm

Rule _____

Working _____

Answer _____

3 Limits

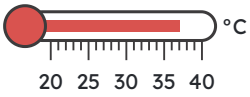


Formula _____

Substitute _____

Answer _____

4 Limits

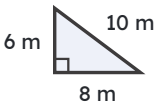


Formula _____

Substitute _____

Answer _____

5 Area Year 8

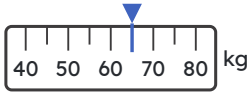


Formula _____

Substitute _____

Answer _____

6 Limits



Formula _____

Substitute _____

Answer _____

7 Convert 10.1



= ? min

Rule _____

Working _____

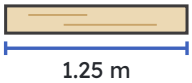
Answer _____

8 These are the limits of accuracy of a length. Find the length and its absolute error.



Answer _____

9 Could this plank really be 1.256 m long? Explain.



Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A nurse's thermometer is marked every 0.4 °C. It shows 37.2 °C. A fever is 37.5 °C or above. Could the patient have a fever? Explain.

Answer _____

2 Two planks are each 1.5 m long, to the nearest 0.1 m. They are laid end to end. Find the smallest and the largest possible total length.

Answer _____

3 Sam ran 100 m in 13.4 s, timed to the nearest 0.1 s. Ali ran it in 13.36 s. Can we be sure that Ali was faster? Explain.

Answer _____

10.3 Percentage error

EASY

Learning intention We are learning to find percentage errors and use them to compare accuracy.

Success criteria
- ☐ I can find the absolute error: half the precision
- ☐ I can calculate percentage error = E ÷ M × 100%
- ☐ I can decide which measurement is more accurate

NOTES Copy what your teacher writes on the board. Date _____

WE DO With your teacher: copy what they write on the board.

Example 1 Find each absolute error or percentage error, to one decimal place where needed.

a Absolute error



E = ?

b % error



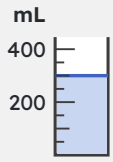
E = 0.5 m
% error = ?

c % error



% error = ?

d % error



% error = ?

YOU DO On your own: work down the box, one round at a time. Show your working. 6 QUESTIONS

REMEMBER p: the place value of the last digit (3.6 kg → 0.1 kg) · E = p ÷ 2 · % error = E ÷ M × 100

ROUND 1 Questions 1–3: find the absolute error E of each measurement.

1



E = ?
nearest m

Formula E = p ÷ 2
Substitute
Answer

2



E = ?

Formula E = p ÷ 2
Substitute
Answer

3

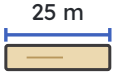


E = ?

Formula E = p ÷ 2
Substitute
Answer

ROUND 2 Questions 4–6: find the percentage error. The absolute error E is given.

4



E = 0.5 m
% error = ?

Formula % error = E ÷ M × 100
Substitute
Answer

5



E = 0.5 kg
% error = ?

Formula % error = E ÷ M × 100
Substitute
Answer

6



E = 0.05 L
% error = ?

Formula % error = E ÷ M × 100
Substitute
Answer

10.3 Percentage error

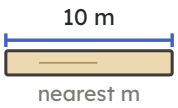
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER E = p ÷ 2 · % error = E ÷ M × 100 · On a scale, p = the gap between two marks · Round to one decimal place.

ROUND 3 Questions 7–9: find the percentage error. Find E first.

7



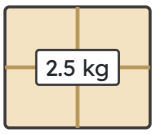
% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

8



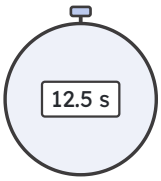
% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

9



% error = ?

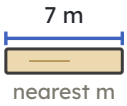
Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

ROUND 4 Questions 10–12: write the formula, then find the percentage error.

10



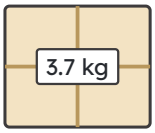
% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

11



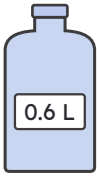
% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

12



% error = ?

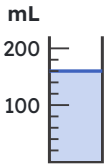
Formula _____

E = p ÷ 2 _____

Answer _____

ROUND 5 Questions 13–15: read the scale, then find the percentage error. p is one mark.

13



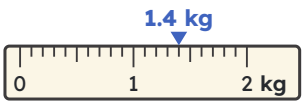
% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

14



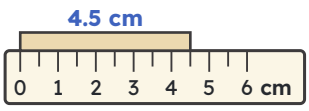
% error = ?

Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

15



% error = ?

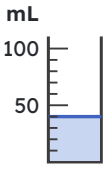
Formula % error = E ÷ M × 100

E = p ÷ 2 _____

Answer _____

ROUND 6 Questions 16–18: write the formula, then find the percentage error.

16



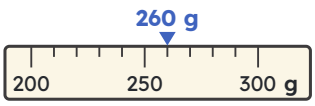
% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

17



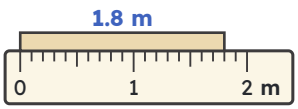
% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

18



% error = ?

Formula _____

E = p ÷ 2 _____

Answer _____

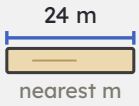
10.3 Percentage error

MEDIUM

WE DO With your teacher: copy what they write on the board.

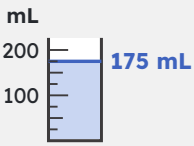
Example 2 Give each answer correct to two decimal places.

a % error

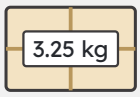


nearest m

b % error



c % error



d Most accurate?

a 24 m

b 175 mL

c 3.25 kg

YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

REMEMBER % error = E ÷ M × 100, to two decimal places · On a scale, p = one mark · Smaller % error: more accurate

1 % error



nearest m

Formula

E = p ÷ 2

Answer

2 Convert 10.1



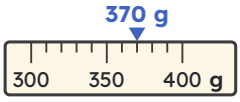
= ? g

Rule

Working

Answer

3 % error



Formula

E = p ÷ 2

Answer

4 % error

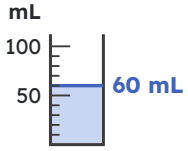


Formula

E = p ÷ 2

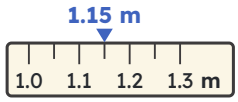
Answer

5 Write the limits of accuracy. 10.2



Answer

6 % error



Formula

E = p ÷ 2

Answer

7 More accurate? **A** 6 m **B** 12 m

nearest m

% error A

% error B

Answer

8 Abs. error 10.2



Formula

Substitute

Answer

9 More accurate? **A** 0.75 kg **B** 2.5 kg

% error A

% error B

Answer

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 Mia's pencil is 15 cm long and she is 168 cm tall, both to the nearest cm. Which measurement is more accurate? Give both percentage errors to two decimal places.

Answer

2 A 100 m sprint is timed as 12.4 s. Find the percentage error of the time, correct to two decimal places.

Answer

3 A jug marked every 50 mL shows 750 mL. Kai says the percentage error is 6.67%. What mistake did he make? Find the correct percentage error.

Answer

10.4 Perimeter and area of composite shapes

EASY

Learning intention We are learning to find the perimeter and area of composite shapes.

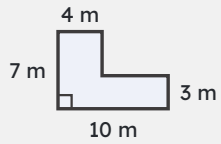
Success criteria
- ☐ I can find missing side lengths, then the perimeter
- ☐ I can find an area by splitting a shape into parts
- ☐ I can find a shaded area by subtracting

NOTES Copy what your teacher writes on the board. **Date** _____

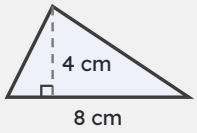
WE DO With your teacher: copy what they write on the board.

Example 1 Find the perimeter or area. Find any missing sides first.

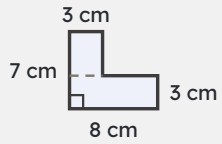
a Perimeter



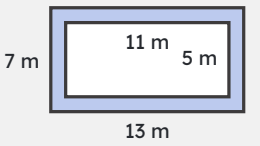
b Area



c Area



d Shaded area

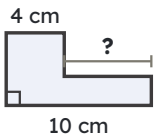


YOU DO On your own: work down the box, one round at a time. Show your working. 6 QUESTIONS

REMEMBER Missing side = whole side − the other part · Perimeter: add every side, the missing ones too.

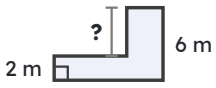
ROUND 1 Questions 1–3: find the missing side.

1



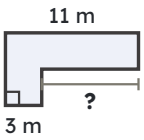
Formula: ? = whole − part
Substitute: _____
Answer: _____

2



Formula: ? = whole − part
Substitute: _____
Answer: _____

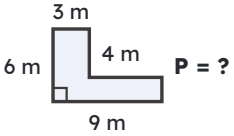
3



Formula: ? = whole − part
Substitute: _____
Answer: _____

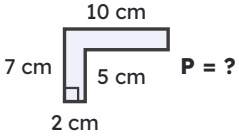
ROUND 2 Questions 4–6: find the perimeter. Find the two missing sides first.

4



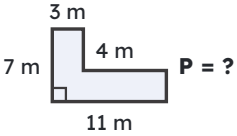
Formula: P = add all sides
Substitute: _____
Answer: _____

5



Formula: P = add all sides
Substitute: _____
Answer: _____

6



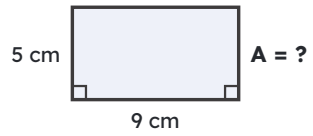
Formula: P = add all sides
Substitute: _____
Answer: _____

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Rectangle A = l × w · Triangle A = ½ × b × h · Split: add the parts · Shaded: whole − hole

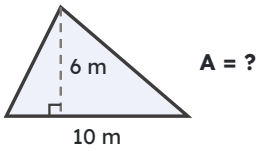
ROUND 3 Questions 7–9: find the area of each rectangle or triangle.

7



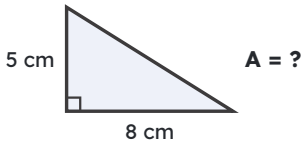
Formula A = l × w
Substitute
Answer

8



Formula A = ½ × b × h
Substitute
Answer

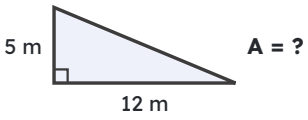
9



Formula A = ½ × b × h
Substitute
Answer

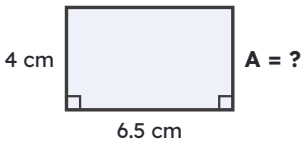
ROUND 4 Questions 10–12: rectangle or triangle? Write the formula, then find the area.

10



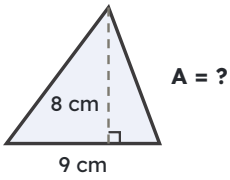
Formula
Substitute
Answer

11



Formula
Substitute
Answer

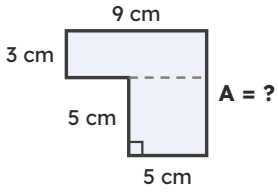
12



Formula
Substitute
Answer

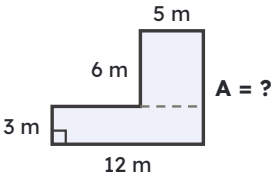
ROUND 5 Questions 13–15: find the area. Split the shape, then add the parts.

13



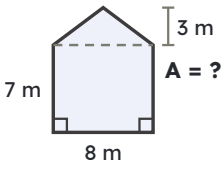
Formula A = l × w + l × w
Substitute
Answer

14



Formula A = l × w + l × w
Substitute
Answer

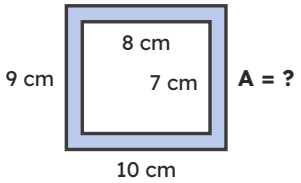
15



Formula A = l × w + ½ × b × h
Substitute
Answer

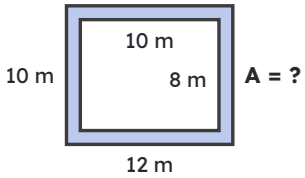
ROUND 6 Questions 16–18: find each shaded area.

16



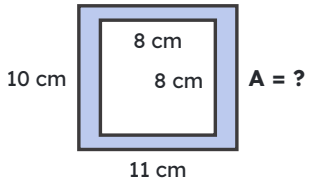
Formula A = whole − hole
Substitute
Answer

17



Formula A = whole − hole
Substitute
Answer

18



Formula A = whole − hole
Substitute
Answer

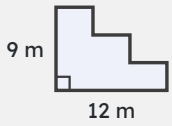
10.4 Perimeter and area of composite shapes

MEDIUM

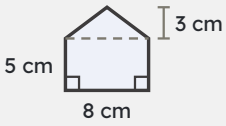
WE DO With your teacher: copy what they write on the board.

Example 2 **Find the perimeter or area. Split the shape, or subtract.**

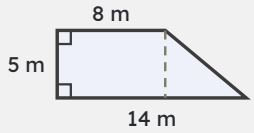
a Perimeter



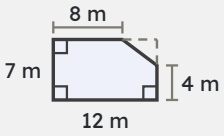
b Area



c Area



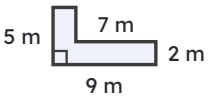
d Area



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

REMEMBER Find missing sides first · Split into rectangles and triangles, or subtract the part cut out

1 Perimeter



Formula _____

Substitute _____

Answer _____

2 In cm [10.1]

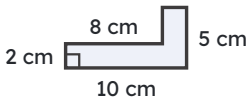


Rule _____

Working _____

Answer _____

3 Area

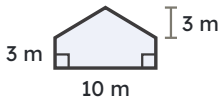


Formula _____

Substitute _____

Answer _____

4 Area



Formula _____

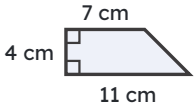
Substitute _____

Answer _____

5 Write the limits of accuracy of 6.4 m. [10.2]

Answer _____

6 Area

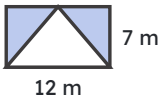


Formula _____

Substitute _____

Answer _____

7 Shaded area

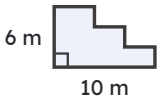


Formula _____

Substitute _____

Answer _____

8 Perimeter



Formula _____

Substitute _____

Answer _____

9 Find the percentage error of 40 cm, measured to the nearest cm. [10.3]

Formula _____

Substitute _____

Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 An L-shaped room is made of a 6 m by 4 m rectangle and a 3 m by 2 m rectangle. Carpet costs \$45 per m². Find the cost to carpet the room.

Answer _____

2 An L-shaped garden bed has outside edges of 8 m and 6 m. Each arm of the L is 2 m wide. Edging costs \$15 per metre. Find the cost to edge the whole bed.

Answer _____

3 The end wall of a shed is a rectangle 4 m wide and 2.5 m high, with a triangle 1.5 m high on top. It has a door 0.9 m wide and 2 m high. Find the area of the wall to paint.

Answer _____

10.5 Areas of quadrilaterals

EASY

Learning intention We are learning to find the area of a parallelogram, a trapezium, a rhombus and a kite.

Success criteria
- ☐ I can use A = bh for a parallelogram
- ☐ I can use A = ½(a + b)h for a trapezium
- ☐ I can use A = ½xy for a rhombus or a kite

NOTES

Copy what your teacher writes on the board.

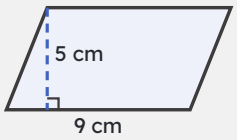
Date _____

WE DO

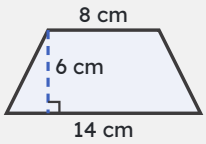
With your teacher: copy what they write on the board.

Example 1 Find the area of each shape. Write the formula first.

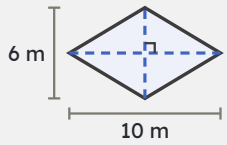
a Parallelogram



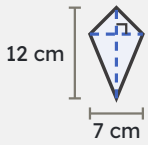
b Trapezium



c Rhombus



d Kite



YOU DO

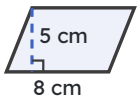
On your own: work down the box, one round at a time. Show your working.

8 QUESTIONS

REMEMBER h is the perpendicular height (the dashed line), at right angles to the base · a and b are the parallel sides.

ROUND 1 Questions 1–4: find the area of each parallelogram.

1



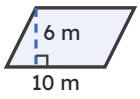
A = ?

Formula A = bh

Substitute _____

Answer _____

2



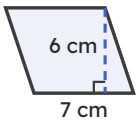
A = ?

Formula A = bh

Substitute _____

Answer _____

3



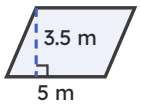
A = ?

Formula A = bh

Substitute _____

Answer _____

4



A = ?

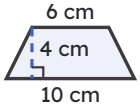
Formula A = bh

Substitute _____

Answer _____

ROUND 2 Questions 5–8: find the area of each trapezium.

5



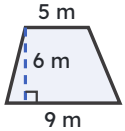
A = ?

Formula A = ½(a + b)h

Substitute _____

Answer _____

6



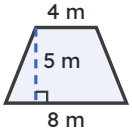
A = ?

Formula A = ½(a + b)h

Substitute _____

Answer _____

7



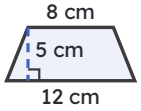
A = ?

Formula A = ½(a + b)h

Substitute _____

Answer _____

8



A = ?

Formula A = ½(a + b)h

Substitute _____

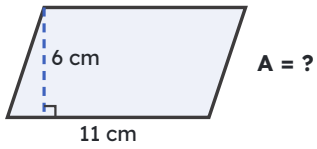
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Parallelogram A = bh · Trapezium A = ½(a + b)h · Rhombus or kite A = ½xy (x and y are the diagonals)

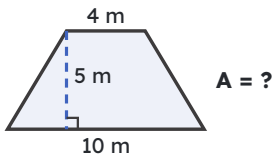
ROUND 3 Questions 9–11: parallelogram or trapezium? Write the formula, then find the area.

9



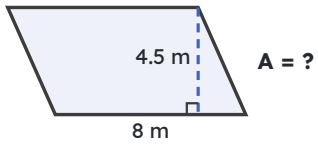
Formula _____
Substitute _____
Answer _____

10



Formula _____
Substitute _____
Answer _____

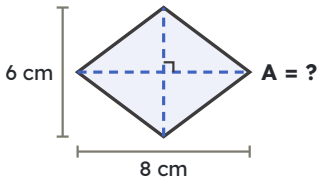
11



Formula _____
Substitute _____
Answer _____

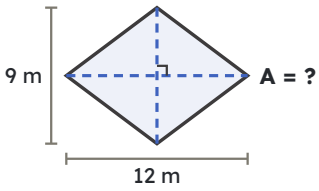
ROUND 4 Questions 12–14: find the area of each rhombus.

12



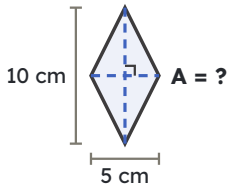
Formula A = ½xy
Substitute _____
Answer _____

13



Formula A = ½xy
Substitute _____
Answer _____

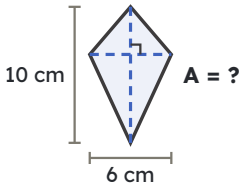
14



Formula A = ½xy
Substitute _____
Answer _____

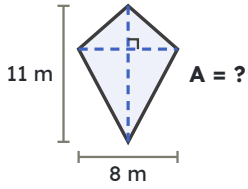
ROUND 5 Questions 15–17: find the area of each kite.

15



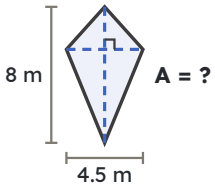
Formula A = ½xy
Substitute _____
Answer _____

16



Formula A = ½xy
Substitute _____
Answer _____

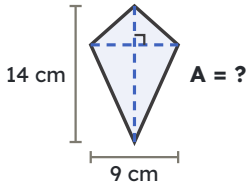
17



Formula A = ½xy
Substitute _____
Answer _____

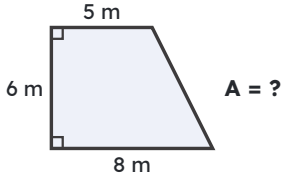
ROUND 6 Questions 18–20: which shape is it? Write the formula, then find the area.

18



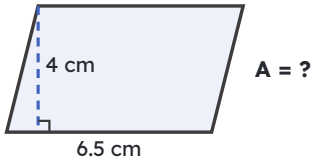
Formula _____
Substitute _____
Answer _____

19



Formula _____
Substitute _____
Answer _____

20

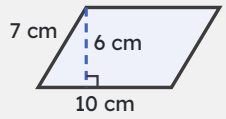


Formula _____
Substitute _____
Answer _____

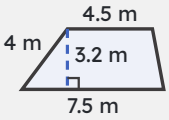
WE DO With your teacher: copy what they write on the board.

Example 2 **Choose the right formula. Some lengths are not needed.**

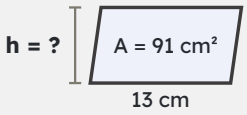
a Area



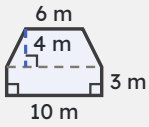
b Area



c Height



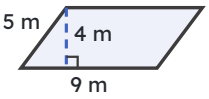
d Area



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

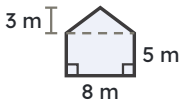
REMEMBER The slanted side is not the height · Missing h: substitute, then solve · Composite: add the parts

1 Area



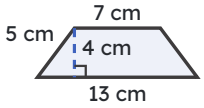
Formula _____
Substitute _____
Answer _____

2 Area 10.4



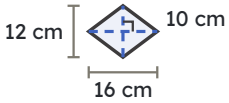
Formula _____
Substitute _____
Answer _____

3 Area



Formula _____
Substitute _____
Answer _____

4 Area

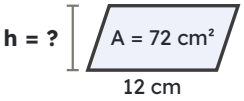


Formula _____
Substitute _____
Answer _____

5 Find the percentage error of 25 kg, measured to the nearest kg. 10.3

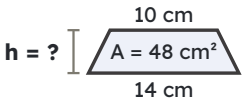
Formula _____
Substitute _____
Answer _____

6 Height



Formula _____
Substitute _____
Answer _____

7 Height

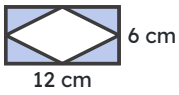


Formula _____
Substitute _____
Answer _____

8 Write the limits of accuracy of 8.5 m. 10.2

Answer _____

9 Shaded area



Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A courtyard is a parallelogram with base 8.5 m and perpendicular height 6 m. Pavers cost $42 per m². Find the cost of paving the courtyard.

Answer _____

2 A kite is made from two rods, 80 cm and 50 cm long, that cross at right angles. Find the area of material needed to cover it.

Answer _____

3 A block of land is a trapezium with parallel sides of 30 m and 45 m. Its area is 750 m². Find the distance between the parallel sides.

Answer _____

10.6 Circumference and arc length

EASY

Learning intention We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use $C = \pi d$ or $C = 2\pi r$ to find a circumference
- ☐ I can find an arc length: $\theta \div 360 \times 2\pi r$
- ☐ I can find the perimeter of sectors and composite shapes

NOTES

Copy what your teacher writes on the board.

Date _____

WE DO

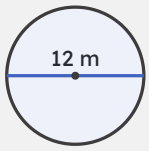
With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

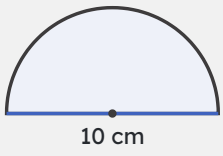
a Circumference



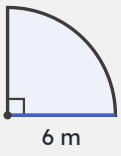
b Circumference



c Perimeter



d Perimeter



YOU DO

On your own: work down the box, one round at a time. Show your working.

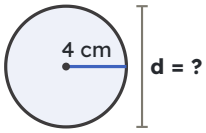
8 QUESTIONS

REMEMBER $d = 2r \cdot r = d \div 2 \cdot C = \pi d \cdot$ Use the π key, then round to one decimal place.

ROUND 1

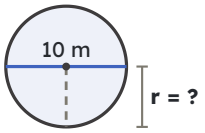
Questions 1–4: find the diameter or the radius.

1



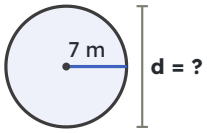
Formula d = 2r
Substitute _____
Answer _____

2



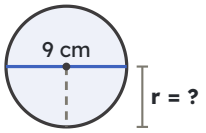
Formula r = d ÷ 2
Substitute _____
Answer _____

3



Formula d = 2r
Substitute _____
Answer _____

4

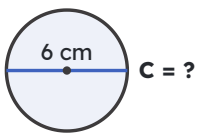


Formula r = d ÷ 2
Substitute _____
Answer _____

ROUND 2

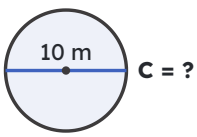
Questions 5–8: find the circumference. The diameter is given.

5



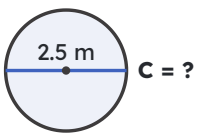
Formula C = πd
Substitute _____
Answer _____

6



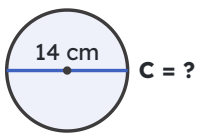
Formula C = πd
Substitute _____
Answer _____

7



Formula C = πd
Substitute _____
Answer _____

8



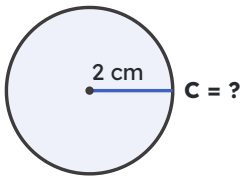
Formula C = πd
Substitute _____
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER C = 2πr = πd · Semicircle P = ½πd + d · Quadrant P = ¼ × 2πr + 2r · Find d first when r is given.

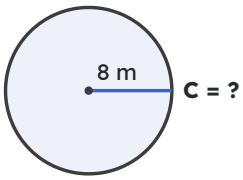
ROUND 3 Questions 9–11: find the circumference. The radius is given.

9



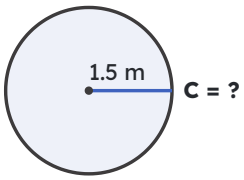
Formula C = 2πr
Substitute
Answer

10



Formula C = 2πr
Substitute
Answer

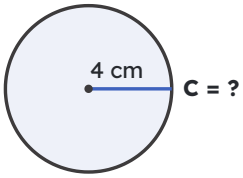
11



Formula C = 2πr
Substitute
Answer

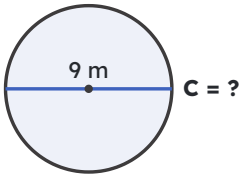
ROUND 4 Questions 12–14: is it r or d? Write the right formula, then find the circumference.

12



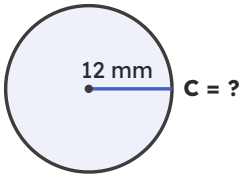
Formula
Substitute
Answer

13



Formula
Substitute
Answer

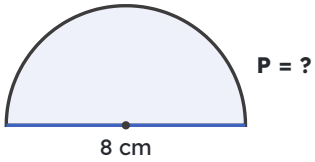
14



Formula
Substitute
Answer

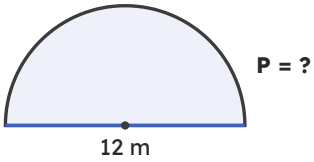
ROUND 5 Questions 15–17: find the perimeter of each semicircle.

15



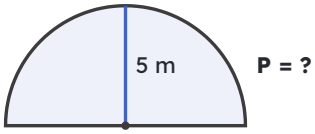
Formula P = ½πd + d
Substitute
Answer

16



Formula P = ½πd + d
Substitute
Answer

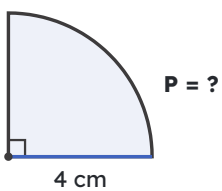
17



Formula P = ½πd + d
Substitute
Answer

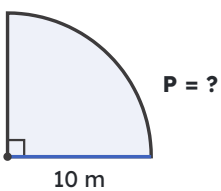
ROUND 6 Questions 18–20: find the perimeter of each quadrant.

18



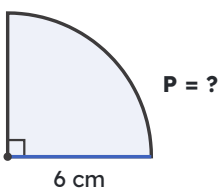
Formula P = ¼ × 2πr + 2r
Substitute
Answer

19



Formula P = ¼ × 2πr + 2r
Substitute
Answer

20



Formula P = ¼ × 2πr + 2r
Substitute
Answer

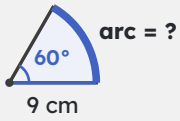
10.6 Circumference and arc length

MEDIUM

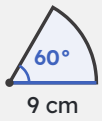
WE DO With your teacher: copy what they write on the board.

Example 2 **Give each answer correct to one decimal place.**

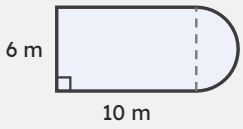
a Arc length



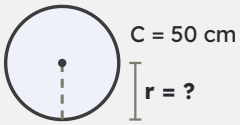
b Perimeter



c Perimeter



d Radius



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

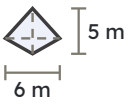
REMEMBER Arc = θ ÷ 360 × 2πr (θ = angle at the centre) · Sector P = arc + 2r · Composite: add the outside edges only

1 Arc length



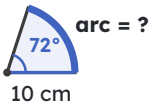
Formula _____
Substitute _____
Answer _____

2 Area 10.5



Formula _____
Substitute _____
Answer _____

3 Arc length



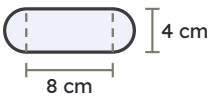
Formula _____
Substitute _____
Answer _____

4 Perimeter



Formula _____
Substitute _____
Answer _____

5 Perimeter



Formula _____
Substitute _____
Answer _____

6 Perimeter

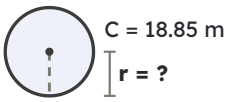


Formula _____
Substitute _____
Answer _____

7 Write the limits of accuracy of 4.5 m. 10.2

Answer _____

8 Radius



Formula _____
Substitute _____
Answer _____

9 Find the percentage error of 20 cm, measured to the nearest cm. 10.3

Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A round garden bed has a diameter of 3.2 m. Edging costs $12 per metre. Find the cost to edge the bed, to the nearest dollar.

Answer _____

2 A bike wheel has a radius of 33 cm. How many full turns does it make to travel 1 km?

Answer _____

3 A running track is a rectangle 84 m long with a semicircle on each end. Each semicircle has a diameter of 64 m. Find one lap, to the nearest metre.

Answer _____

10.7 Areas of circles and sectors

EASY

Learning intention We are learning to find the areas of circles, semicircles, quadrants, sectors and composite shapes.

Success criteria
- ☐ I can use $A = \pi r^2$ to find the area of a circle
- ☐ I can find the area of a sector: $\theta \div 360 \times \pi r^2$
- ☐ I can find composite and shaded areas

NOTES Copy what your teacher writes on the board. **Date** _____

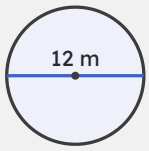
WE DO With your teacher: copy what they write on the board.

Example 1 Find the area of each shape, correct to one decimal place.

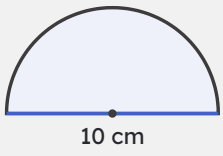
a Area



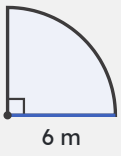
b Area



c Area



d Area



YOU DO On your own: work down the box, one round at a time. Show your working. 8 QUESTIONS

REMEMBER $A = \pi r^2$ · $r = d \div 2$ · Use the π key, then round to one decimal place · Area units: cm^2, m^2.

ROUND 1 Questions 1–4: find the area of each circle. The radius is given.

1



A = ?

Formula $A = \pi r^2$

Substitute _____

Answer _____

2



A = ?

Formula $A = \pi r^2$

Substitute _____

Answer _____

3



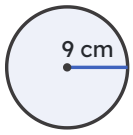
A = ?

Formula $A = \pi r^2$

Substitute _____

Answer _____

4



A = ?

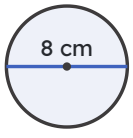
Formula $A = \pi r^2$

Substitute _____

Answer _____

ROUND 2 Questions 5–8: find the area. The diameter is given: halve it first.

5



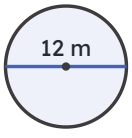
A = ?

Formula $A = \pi r^2$

Substitute _____

Answer _____

6



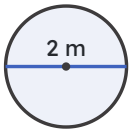
A = ?

Formula $A = \pi r^2$

Substitute _____

Answer _____

7



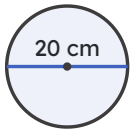
A = ?

Formula $A = \pi r^2$

Substitute _____

Answer _____

8



A = ?

Formula $A = \pi r^2$

Substitute _____

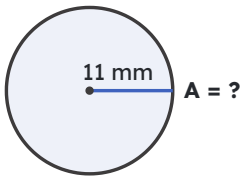
Answer _____

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER $A = \pi r^2$ · Semicircle $A = \frac{1}{2}\pi r^2$ · Quadrant $A = \frac{1}{4}\pi r^2$ · Sector $A = \theta \div 360 \times \pi r^2$ · Halve d first.

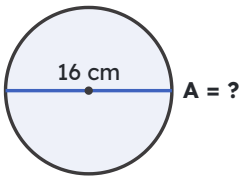
ROUND 3 Questions 9–11: is it r or d? Write the formula, then find the area.

9



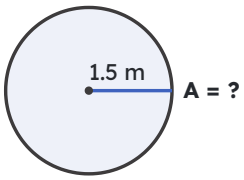
Formula _____
Substitute _____
Answer _____

10



Formula _____
Substitute _____
Answer _____

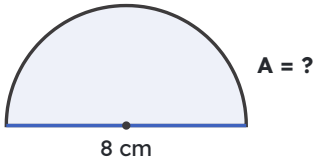
11



Formula _____
Substitute _____
Answer _____

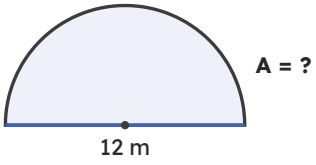
ROUND 4 Questions 12–14: find the area of each semicircle.

12



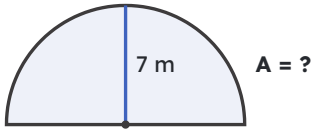
Formula $A = \frac{1}{2}\pi r^2$
Substitute _____
Answer _____

13



Formula $A = \frac{1}{2}\pi r^2$
Substitute _____
Answer _____

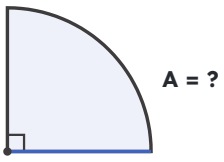
14



Formula $A = \frac{1}{2}\pi r^2$
Substitute _____
Answer _____

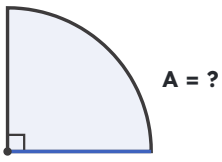
ROUND 5 Questions 15–17: find the area of each quadrant.

15



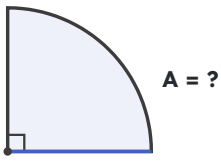
Formula $A = \frac{1}{4}\pi r^2$
Substitute _____
Answer _____

16



Formula $A = \frac{1}{4}\pi r^2$
Substitute _____
Answer _____

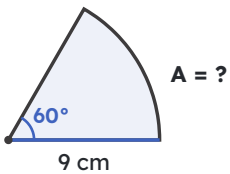
17



Formula $A = \frac{1}{4}\pi r^2$
Substitute _____
Answer _____

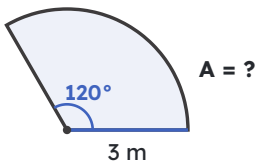
ROUND 6 Questions 18–20: find the area of each sector.

18



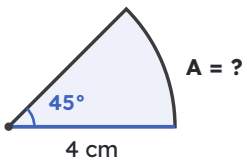
Formula $A = \theta \div 360 \times \pi r^2$
Substitute _____
Answer _____

19



Formula $A = \theta \div 360 \times \pi r^2$
Substitute _____
Answer _____

20



Formula $A = \theta \div 360 \times \pi r^2$
Substitute _____
Answer _____

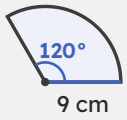
10.7 Areas of circles and sectors

MEDIUM

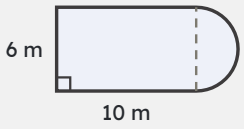
WE DO With your teacher: copy what they write on the board.

Example 2 **Give each answer correct to one decimal place.**

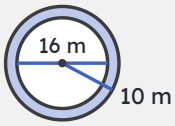
a Area



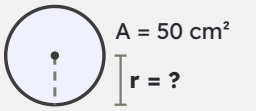
b Area



c Shaded area



d Radius



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

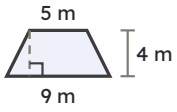
REMEMBER Sector A = θ ÷ 360 × πr² · Composite: add the parts · Shaded: whole − cut-out · Radius: r² = A ÷ π, then √

1 Area



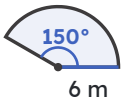
Formula _____
Substitute _____
Answer _____

2 Area 10.5



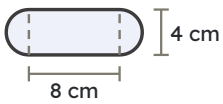
Formula _____
Substitute _____
Answer _____

3 Area



Formula _____
Substitute _____
Answer _____

4 Area



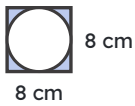
Formula _____
Substitute _____
Answer _____

5 Perimeter 10.6



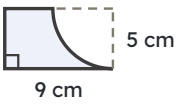
Formula _____
Substitute _____
Answer _____

6 Shaded area



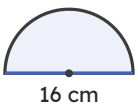
Formula _____
Substitute _____
Answer _____

7 Area



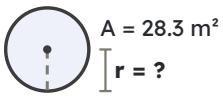
Formula _____
Substitute _____
Answer _____

8 Perimeter 10.6



Formula _____
Substitute _____
Answer _____

9 Radius



Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A garden sprinkler sprays water 6 m and turns through an angle of 120°. Find the area of lawn it waters.

Answer _____

2 A round pond has a radius of 3 m. A path 1 m wide goes all the way around it. Find the area of the path.

Answer _____

3 A pizza has a diameter of 30 cm. It is cut into 8 equal slices. Find the area of one slice.

Answer _____

10.8 Surface area of a prism

EASY

Learning intention We are learning to find the surface area of a prism by adding the areas of its faces.

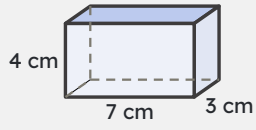
Success criteria
- ☐ I can find the area of each face of a prism
- ☐ I can use SA = $6s^2$ and SA = 2(lw + lh + wh)
- ☐ I can find the surface area of triangular prisms and open boxes

NOTES Copy what your teacher writes on the board. **Date** _____

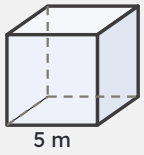
WE DO With your teacher: copy what they write on the board.

Example 1 Find each area. Surface area (SA) is the total area of all the faces.

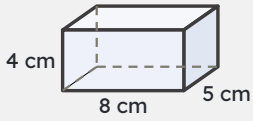
a Shaded face



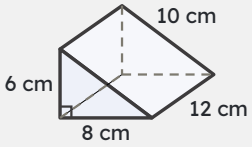
b Surface area



c Surface area



d Surface area

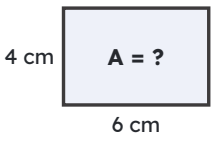


YOU DO On your own: work down the box, one round at a time. Show your working. 7 QUESTIONS

REMEMBER Each face is a rectangle: A = l × w · Front: l × h · Top: l × w · Side: w × h · Areas are in square units (cm²).

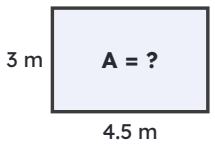
ROUND 1 Questions 1–4: find the area of each face.

1



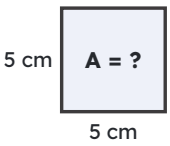
Formula A = l × w
Substitute _____
Answer _____

2



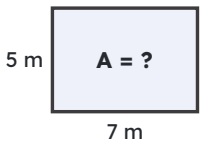
Formula A = l × w
Substitute _____
Answer _____

3



Formula A = l × w
Substitute _____
Answer _____

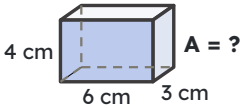
4



Formula A = l × w
Substitute _____
Answer _____

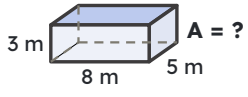
ROUND 2 Questions 5–7: find the area of the shaded face of each rectangular prism.

5



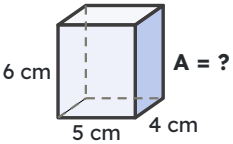
Formula A = l × h
Substitute _____
Answer _____

6



Formula A = l × w
Substitute _____
Answer _____

7



Formula A = w × h
Substitute _____
Answer _____

10.8 Surface area of a prism

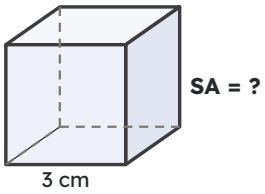
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Cube: SA = $6s^2$ · Rectangular prism: SA = 2(lw + lh + wh) · Triangular prism: 2 triangles + 3 rectangles

ROUND 3 Questions 8–10: find the surface area of each cube.

8

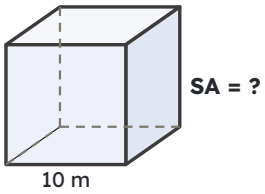


Formula SA = $6s^2$

Substitute

Answer

9

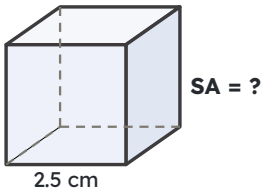


Formula SA = $6s^2$

Substitute

Answer

10



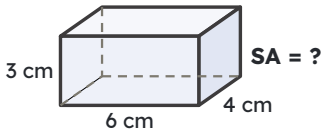
Formula SA = $6s^2$

Substitute

Answer

ROUND 4 Questions 11–13: find the surface area of each rectangular prism.

11

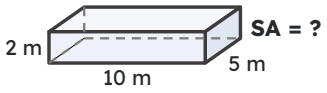


Formula SA = 2(lw + lh + wh)

Substitute

Answer

12

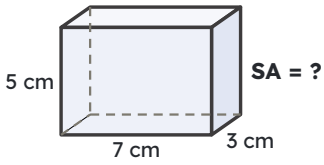


Formula SA = 2(lw + lh + wh)

Substitute

Answer

13



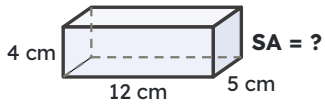
Formula SA = 2(lw + lh + wh)

Substitute

Answer

ROUND 5 Questions 14–16: cube or rectangular prism? Write the formula, then find the surface area.

14

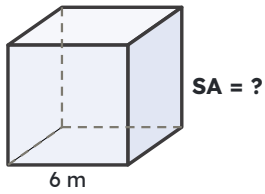


Formula

Substitute

Answer

15

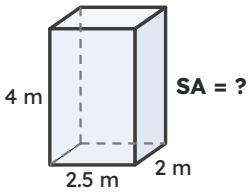


Formula

Substitute

Answer

16



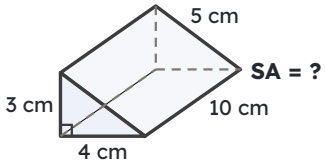
Formula

Substitute

Answer

ROUND 6 Questions 17–19: find the surface area of each right-angled triangular prism.

17

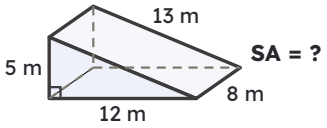


Formula SA = bh + (b + h + s)l

Substitute

Answer

18

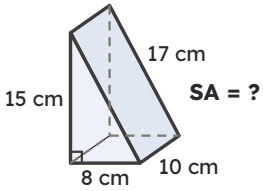


Formula SA = bh + (b + h + s)l

Substitute

Answer

19



Formula SA = bh + (b + h + s)l

Substitute

Answer

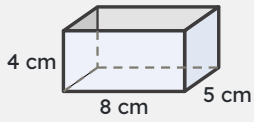
10.8 Surface area of a prism

MEDIUM

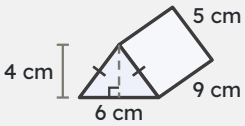
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each surface area. List the faces first.**

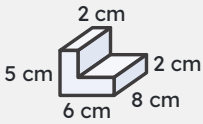
a SA (no lid)



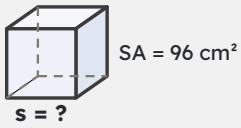
b Surface area



c Surface area



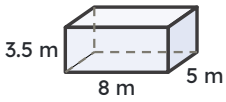
d Side length



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

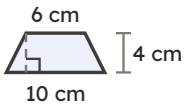
REMEMBER No lid: SA = lw + 2lh + 2wh · Any prism: 2 × end + perimeter × length · Cube: s^2 = SA ÷ 6 · π answers to 1 d.p.

1 Surface area



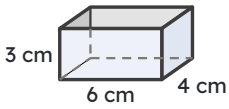
Formula _____
Substitute _____
Answer _____

2 Area 10.5



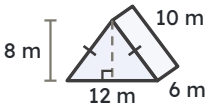
Formula _____
Substitute _____
Answer _____

3 SA (no lid)



Formula _____
Substitute _____
Answer _____

4 Surface area



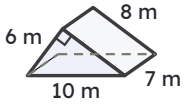
Formula _____
Substitute _____
Answer _____

5 Area 10.7



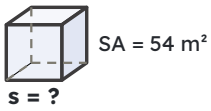
Formula _____
Substitute _____
Answer _____

6 Surface area



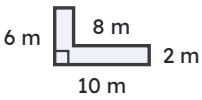
Formula _____
Substitute _____
Answer _____

7 Side length



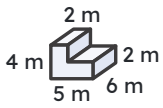
Formula _____
Substitute _____
Answer _____

8 Area 10.4



Formula _____
Substitute _____
Answer _____

9 Surface area



Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A gift box is 30 cm long, 20 cm wide and 10 cm high. Find the area of paper needed to cover it exactly.

Answer _____

2 A room is 5 m long, 4 m wide and 2.5 m high. 1 L of paint covers 10 m². How many litres of paint are needed for the four walls and the ceiling?

Answer _____

3 A tent is a triangular prism 4 m long. Each end is a triangle: base 3 m, height 2 m, sloping sides 2.5 m. Find the area of canvas, including the floor.

Answer _____

10.9 Surface area of a cylinder

EASY

Learning intention We are learning to find the surface area of a cylinder: two circles and a rolled-up rectangle.

Success criteria
- ☐ I can find the area of a circular end: $A = \pi r^2$
- ☐ I can find the curved surface area: $A = 2\pi rh$
- ☐ I can find the surface area of closed and open cylinders

NOTES

Copy what your teacher writes on the board.

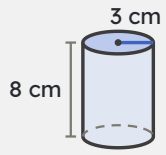
Date _____

WE DO

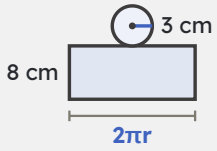
With your teacher: copy what they write on the board.

Example 1 Find each area, correct to one decimal place.

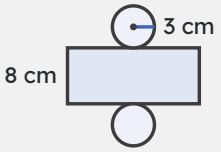
a End area



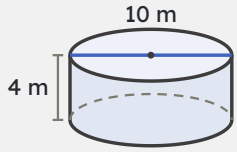
b Curved surface



c Surface area



d Surface area



YOU DO

On your own: work down the box, one round at a time. Show your working.

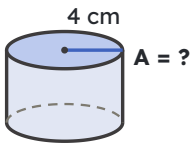
7 QUESTIONS

REMEMBER End: $A = \pi r^2$ · Curved surface: $A = 2\pi rh$ (h is the height) · Use the π key · Round to one decimal place.

ROUND 1

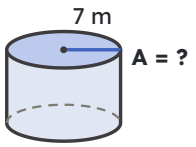
Questions 1–4: find the area of the shaded end of each cylinder.

1



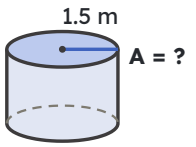
Formula $A = \pi r^2$
Substitute _____
Answer _____

2



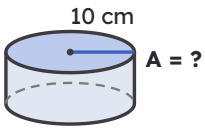
Formula $A = \pi r^2$
Substitute _____
Answer _____

3



Formula $A = \pi r^2$
Substitute _____
Answer _____

4

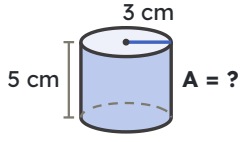


Formula $A = \pi r^2$
Substitute _____
Answer _____

ROUND 2

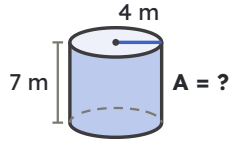
Questions 5–7: find the curved surface area (shaded).

5



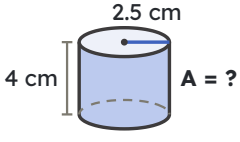
Formula $A = 2\pi rh$
Substitute _____
Answer _____

6



Formula $A = 2\pi rh$
Substitute _____
Answer _____

7



Formula $A = 2\pi rh$
Substitute _____
Answer _____

10.9 Surface area of a cylinder

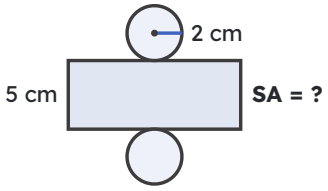
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Closed cylinder: SA = 2πr² + 2πrh (two ends + the curved surface) · r = d ÷ 2 · Round to one decimal place.

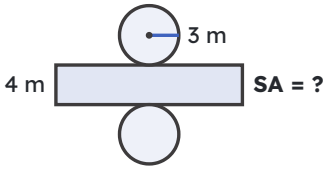
ROUND 3 Questions 8–10: find the total area of each net. It is the surface area of the cylinder.

8



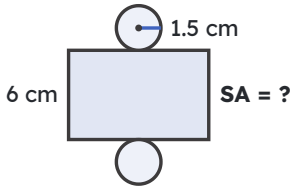
Formula SA = 2πr² + 2πrh
Substitute
Answer

9



Formula SA = 2πr² + 2πrh
Substitute
Answer

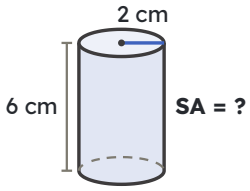
10



Formula SA = 2πr² + 2πrh
Substitute
Answer

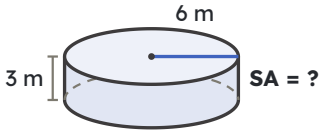
ROUND 4 Questions 11–13: find the surface area of each closed cylinder. The radius is given.

11



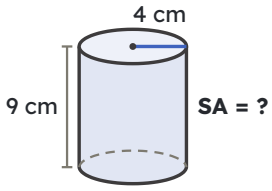
Formula SA = 2πr² + 2πrh
Substitute
Answer

12



Formula SA = 2πr² + 2πrh
Substitute
Answer

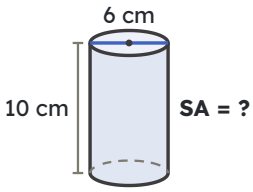
13



Formula SA = 2πr² + 2πrh
Substitute
Answer

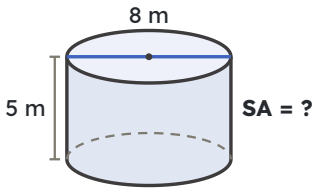
ROUND 5 Questions 14–16: find the surface area. The diameter is given: halve it first.

14



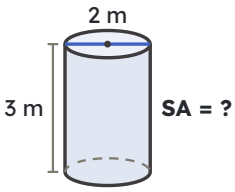
Formula SA = 2πr² + 2πrh
Substitute
Answer

15



Formula SA = 2πr² + 2πrh
Substitute
Answer

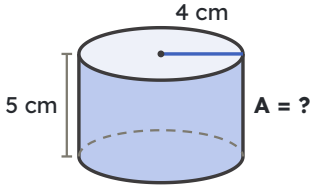
16



Formula SA = 2πr² + 2πrh
Substitute
Answer

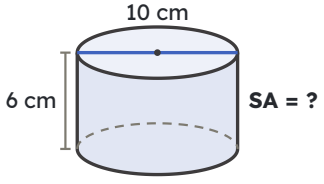
ROUND 6 Questions 17–19: which surface? Write the formula, then find the area.

17



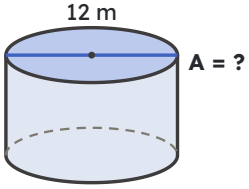
Formula
Substitute
Answer

18



Formula
Substitute
Answer

19



Formula
Substitute
Answer

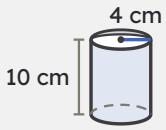
10.9 Surface area of a cylinder

MEDIUM

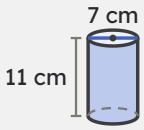
WE DO With your teacher: copy what they write on the board.

Example 2 **Find each area, correct to one decimal place. List the surfaces first.**

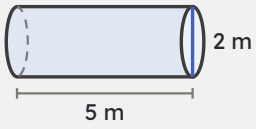
a SA (no lid)



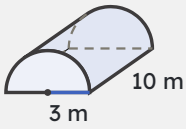
b Label area



c Surface area



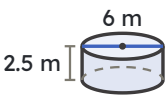
d Surface area



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

REMEMBER No lid: SA = $\pi r^2 + 2\pi rh$ · Label: $2\pi rh$ only · Half cylinder: SA = $\pi r^2 + \pi rh + 2rh$ · Lying down: h is the length

1 Surface area



Formula _____
Substitute _____
Answer _____

2 Perimeter 10.6



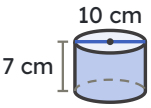
Formula _____
Substitute _____
Answer _____

3 SA (no lid)



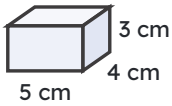
Formula _____
Substitute _____
Answer _____

4 Label area



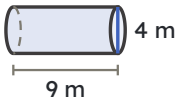
Formula _____
Substitute _____
Answer _____

5 SA 10.8



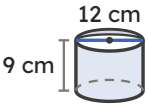
Formula _____
Substitute _____
Answer _____

6 Surface area



Formula _____
Substitute _____
Answer _____

7 SA (no lid)



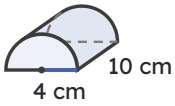
Formula _____
Substitute _____
Answer _____

8 Area 10.7



Formula _____
Substitute _____
Answer _____

9 Surface area



Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A closed cylindrical tank has a diameter of 2 m and a height of 1.5 m. Paint costs \$14 per m². Find the cost to paint the whole tank, to the nearest dollar.

Answer _____

2 A paint roller is a cylinder 25 cm long with a diameter of 6 cm. Find the area it paints in 10 full turns.

Answer _____

3 A bucket with no lid is a cylinder with a diameter of 30 cm and a height of 25 cm. Find the area of plastic needed to make it.

Answer _____

10.10 Volumes of prisms and cylinders

EASY

Learning intention We are learning to find the volume of a prism or a cylinder ($V = Ah$), and its capacity.

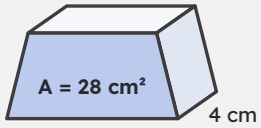
Success criteria
- ☐ I can use $V = Ah$ to find the volume of any prism
- ☐ I can use $V = \pi r^2 h$ to find the volume of a cylinder
- ☐ I can change between cm^3, mL, L and m^3

NOTES Copy what your teacher writes on the board. **Date** _____

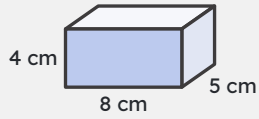
WE DO With your teacher: copy what they write on the board.

Example 1 Find the volume of each solid. Give the cylinder correct to one decimal place.

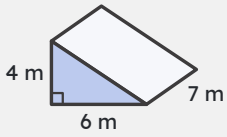
a Prism



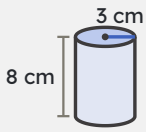
b Cuboid



c Triangular prism



d Cylinder

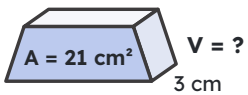


YOU DO On your own: work down the box, one round at a time. Show your working. 6 QUESTIONS

REMEMBER $V = Ah$: A is the area of the shaded end, h the distance between the ends · Cuboid: $V = lwh$ · Units: cm^3, m^3

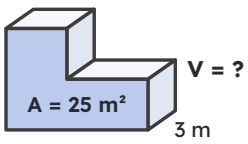
ROUND 1 Questions 1–3: find the volume. The area of the shaded end is given.

1



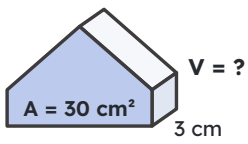
Formula V = Ah
Substitute _____
Answer _____

2



Formula V = Ah
Substitute _____
Answer _____

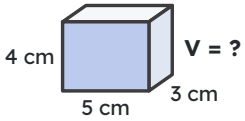
3



Formula V = Ah
Substitute _____
Answer _____

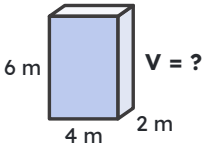
ROUND 2 Questions 4–6: find the volume of each cuboid.

4



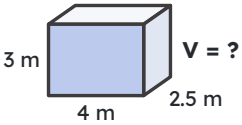
Formula V = lwh
Substitute _____
Answer _____

5



Formula V = lwh
Substitute _____
Answer _____

6



Formula V = lwh
Substitute _____
Answer _____

10.10 Volumes of prisms and cylinders

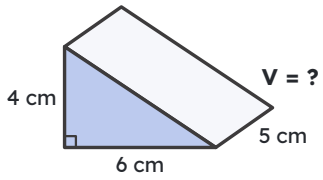
EASY

YOU DO On your own: keep going, one round at a time. Show your working. 12 QUESTIONS

REMEMBER Triangular prism: V = ½bh × l · Cylinder: V = πr²h · r = d ÷ 2 · Use the π key, then round to one decimal place.

ROUND 3 Questions 7–9: find the volume of each triangular prism.

7

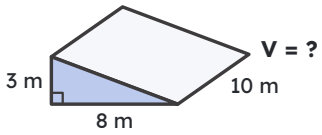


Formula V = ½bh × l

Substitute

Answer

8

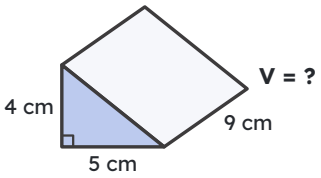


Formula V = ½bh × l

Substitute

Answer

9



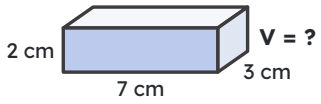
Formula V = ½bh × l

Substitute

Answer

ROUND 4 Questions 10–12: which solid is it? Write its formula, then find the volume.

10

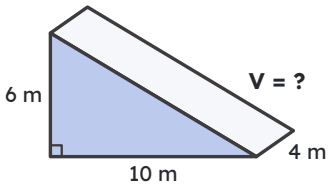


Formula

Substitute

Answer

11

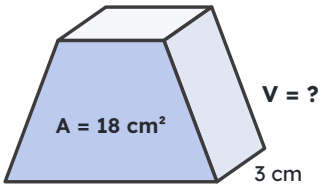


Formula

Substitute

Answer

12



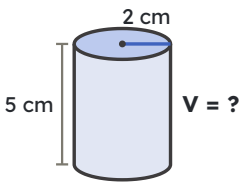
Formula

Substitute

Answer

ROUND 5 Questions 13–15: find the volume of each cylinder. The radius is given.

13

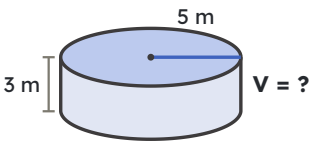


Formula V = πr²h

Substitute

Answer

14

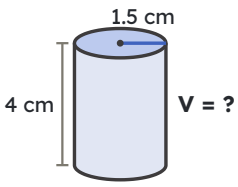


Formula V = πr²h

Substitute

Answer

15



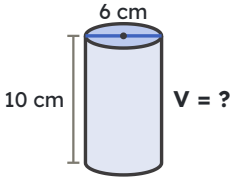
Formula V = πr²h

Substitute

Answer

ROUND 6 Questions 16–18: find the volume. The diameter is given: halve it first.

16

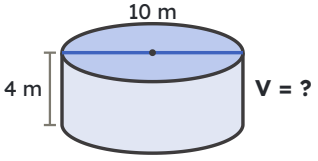


Formula V = πr²h

Substitute

Answer

17

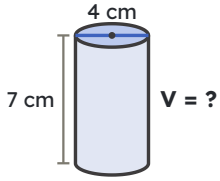


Formula V = πr²h

Substitute

Answer

18



Formula V = πr²h

Substitute

Answer

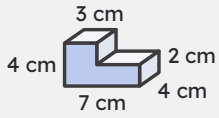
10.10 Volumes of prisms and cylinders

MEDIUM

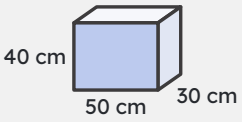
WE DO With your teacher: copy what they write on the board.

Example 2 Round π answers to one decimal place. 1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L.

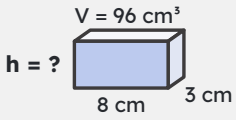
a Volume



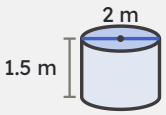
b Capacity in L



c Height



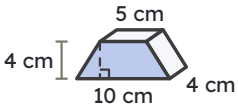
d Capacity in L



YOU DO On your own: work in order. A tag shows an earlier lesson. Show your working. 9 QUESTIONS

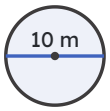
REMEMBER Find A, the end, first · L-shape: two rectangles · Trapezium: A = ½(a + b)h · Height: h = V ÷ (l × w)

1 Volume



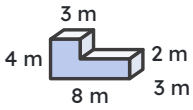
Formula _____
Substitute _____
Answer _____

2 Area 10.7



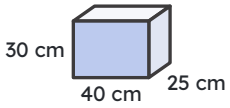
Formula _____
Substitute _____
Answer _____

3 Volume



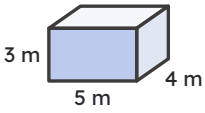
Formula _____
Substitute _____
Answer _____

4 Capacity in L



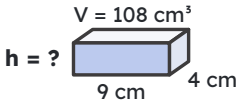
Formula _____
Substitute _____
Answer _____

5 Surface area 10.8



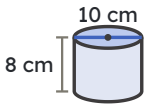
Formula _____
Substitute _____
Answer _____

6 Height



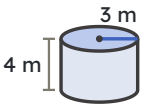
Formula _____
Substitute _____
Answer _____

7 Capacity in mL



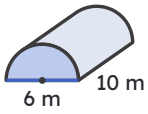
Formula _____
Substitute _____
Answer _____

8 Surface area 10.9



Formula _____
Substitute _____
Answer _____

9 Volume



Formula _____
Substitute _____
Answer _____

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working. 3 PROBLEMS

1 A tent is a triangular prism. Its triangular end has a base of 2.4 m and a height of 1.5 m, and the tent is 3 m long. Find the volume of air inside the tent.

Answer _____

2 A cylindrical rainwater tank has a diameter of 3 m and a height of 2 m. How many litres does it hold, to the nearest litre?

Answer _____

3 A pool is 25 m long and 10 m wide. It holds 375 000 L of water. How deep is the water?

Answer _____